

SIEM Visualization Development (HACK THE BOX)

For this exercise report we will be using Elastic stack to develop visual dashboards on HacktheBox.

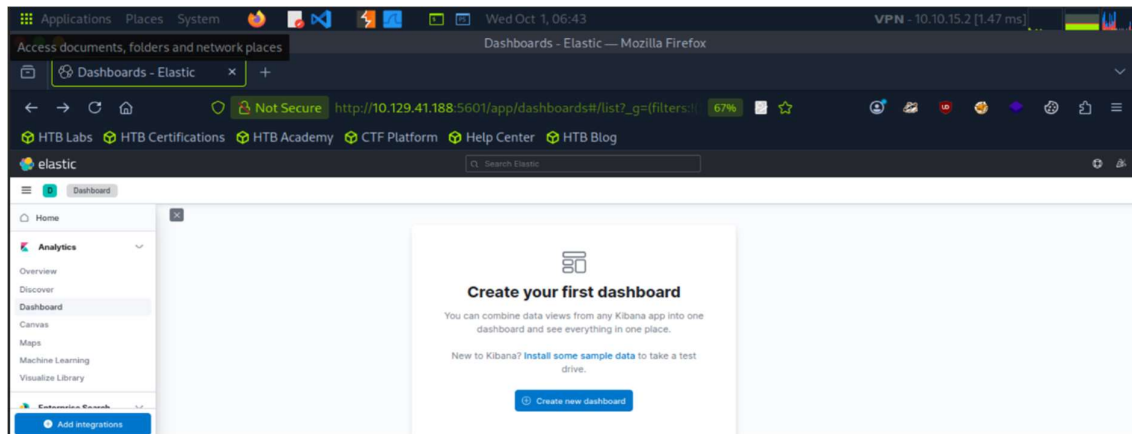
We will create the Visualization for following examples:

- 1) Example 1: Failed Logon Attempts (All Users)
- 2) Example 2: Failed Logon Attempts (Disabled Users)
- 3) Example 3: Successful RDP Logon Related to Service Accounts
- 4) Example 4: User Added or Removed from a Local Group

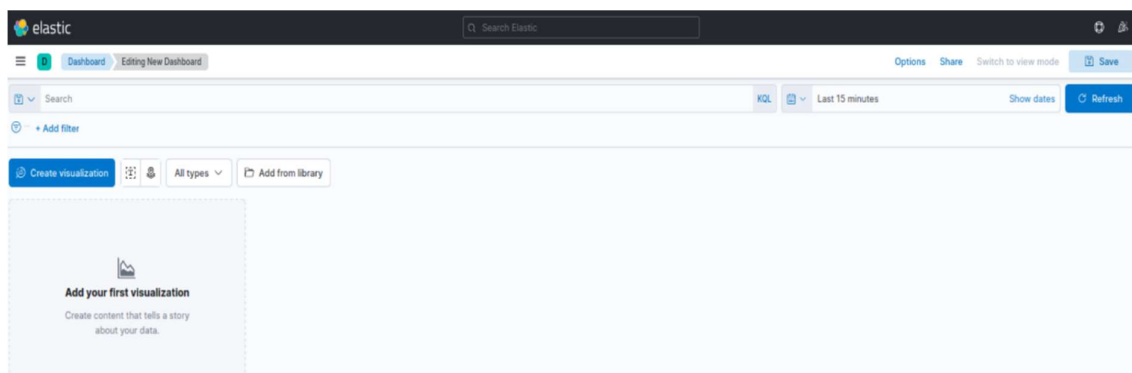
Example 1: Failed Logon Attempts (All Users)

Created a SIEM dashboard visualization to monitor and analyze failed Windows logon attempts, providing Security Operations Center (SOC) analysts with actionable security intelligence.

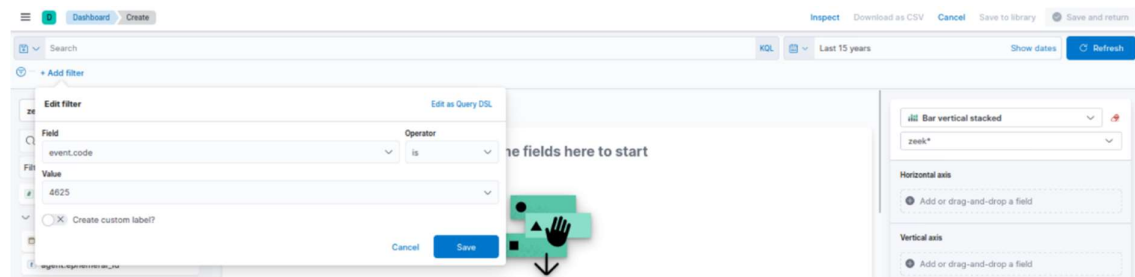
I started by logging into my Elasticstack SIEM and selecting the dashboard from navigation pane.



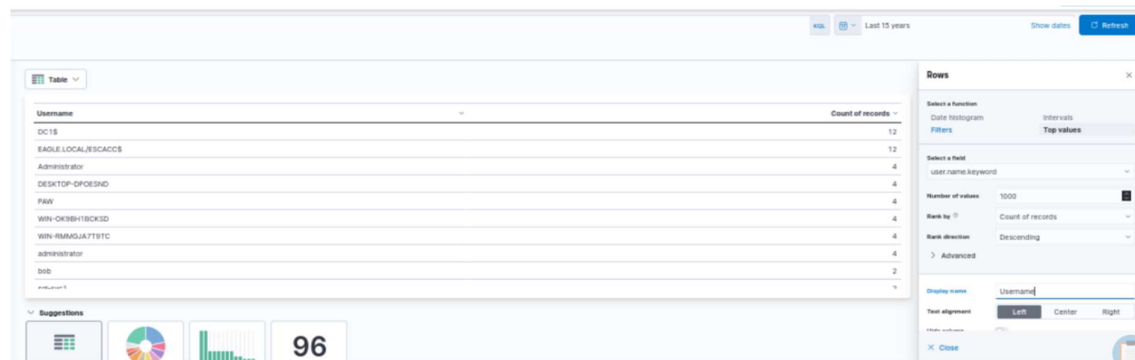
We will click on “Create New Dashboard”



Now we click **“Create Visualization”**



Click **“add filter”** and write **event.code** and value **4625 (ID for Failed Login Attempt)**



Now we filter the **Rows**:

Configure Table Rows

- **First Row:** Field user.name.keyword → Display Name: Username
 - Size: 1000 (show all unique users)



- **Second Row:** Field host.hostname.keyword → Display Name: Event logged by
 - Size: 1000 (show all affected machines)

This screenshot shows the configuration for a visualization. The main table lists the 'Event logged by' field with its corresponding 'Count of records'. The 'Rows' panel on the right shows the configuration for the visualization, including the field 'Event logged by' and the 'Count of records' metric.

Username	Event logged by	Count of records
DC1S	DC1	10
DC1S	DC2	2
EAGLE.LOCAL/ESCADES	PKI	12
Administrator	DC1	4
DESKTOP-GPOESND	DC1	4
PAW	DC2	4
WIN-OKSBH18CKSD	DC1	4
WIN-RMMSJA779TC	DC1	4
Administrator	PAW	4
Administrator	PKI	1

- **Third Row:** Field winlog.logon.type.keyword → Display Name: Logon Type
 - Size: 1000 (show authentication methods)

This screenshot shows the configuration for a visualization. The main table lists the 'Logon Type' field with its corresponding 'Count of records'. The 'Rows' panel on the right shows the configuration for the visualization, including the field 'Logon Type' and the 'Count of records' metric.

Username	Event logged by	Logon Type	Count of records
DC1S	DC1	Network	10
DC1S	DC2	Network	2
EAGLE.LOCAL/ESCADES	PKI	Network	12
Administrator	DC1	Interactive	3
Administrator	DC1	Unlock	1
DESKTOP-GPOESND	DC1	Network	4
PAW	DC2	Network	4
WIN-OKSBH18CKSD	DC1	Network	4
WIN-RMMSJA779TC	DC1	Network	4
Administrator	ESCADES	Interactive	1

Configure Metrics

- Set metric to **"Count"** of records
- Display Name: # of logins
- Sorted table descending by # of logins to show most suspicious activity first

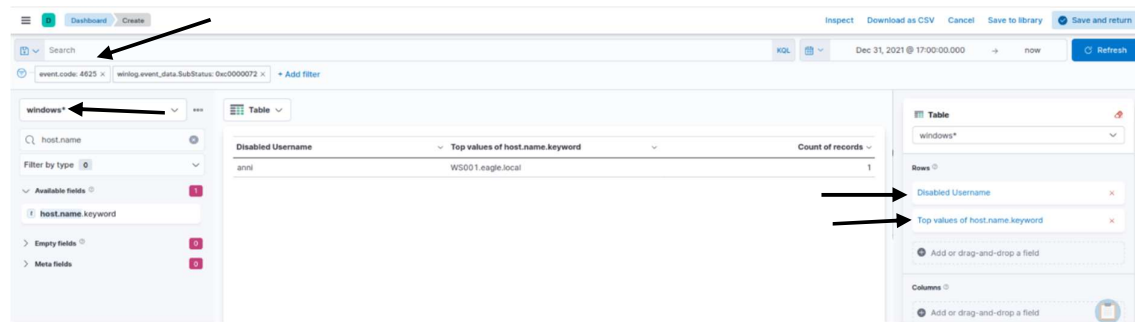
Initial Save

- Saved visualization as **"Failed Logon Attempts"**
- Saved dashboard as **"SOC-Alerts"**

This screenshot shows the configuration for a visualization. The main table lists the 'Logon Type' field with its corresponding '# of Logins'. The 'Rows' panel on the right shows the configuration for the visualization, including the field 'Logon Type' and the '# of Logins' metric.

Username	Event logged by	Logon Type	# of Logins
PAW	DC2	Network	4
Administrator	DC1	Interactive	3
Administrator	PAW	Interactive	2
bob	WS001	Interactive	2
top-exec1	PKI	Network	2
Administrator	DC1	Unlock	1
Administrator	PAW	Unlock	1
Administrator	DC2	Interactive	1
ADMIN	WS001	Interactive	1
Administrator	PKI	Interactive	1

Example 2: Failed Logon Attempts (Disabled User)



Technical Implementation:

- Filter: `event.code: 4625 AND winlog.event_data.SubStatus: 0xc0000072`
- Table visualization showing disabled users targeted and attack sources

Key Learning:

- Understanding Windows SubStatus codes for detailed logon failure analysis
- Detecting attacks where valid credentials are used against disabled accounts
- Building high-priority security alerts for SOC monitoring

Example 3: Successful RDP Logon Related to Service Accounts

1. Filters You Need:

- `event.code: 4624` (Successful logon)
- `winlog.logon.type: RemoteInteractive` (RDP logons)
- KQL: `user.name: svc-*` (Service accounts starting with "svc-")

2. Table Structure:

- **Rows:** `user.name.keyword` (Service account name)
- **Rows:** `host.hostname.keyword` (Target machine)
- **Rows:** `related.ip.keyword` (Source IP that initiated RDP)
- **Metrics:** Count (Number of occurrences)

Real-World Impact

In a real SOC, this visualization would help detect:

- Attackers using stolen service account credentials
- Insiders misusing privileged accounts

- Lateral movement using service accounts

Service Account Name	Target Machines	Source IP that initiated RDP	# of Logins
svc-sql1	PKI	192.168.28.130	2

Example 4: Local Administrator Group Changes

Create a security monitor to track user additions and removals from the local Administrators group, detecting potential privilege escalation and persistence attacks.

Security Significance

- **Attackers often add themselves to local admin groups** for persistence
- **Unauthorized privilege escalation** detection
- **Compliance monitoring** for privileged access changes
- **High-severity alert** in SOC environments

Technical Implementation

Filters Applied:

1. Event Code Filter:

- **Field:** event.code
- **Operator:** is one of
- **Value:** 4732, 4733
- **Purpose:** Capture both user additions (4732) and removals (4733) from local groups

2. Group Name Filter:

- **Field:** group.name
- **Operator:** is
- **Value:** Administrators

- **Purpose:** Focus only on the high-privilege Administrators group

3. Time Range Filter:

- **From:** March 5, 2023
- **To:** Present
- **Purpose:** Monitor changes within specific compliance timeframe

Table Rows Configuration:

- user.name.keyword - User who performed the group change
- winlog.event_data.MemberSid.keyword - User that was added/removed from group
- group.name.keyword - Group name (verifies "Administrators")
- event.action.keyword - Action type ("added" or "removed")
- host.name.keyword - Machine where change occurred
- **Metrics:** Count of records

Key Learning Outcomes

- Understanding Windows security event IDs for group membership changes
- Implementing focused monitoring on privileged groups only
- Using time-based filtering for compliance requirements
- Detecting potential attacker persistence mechanisms

Real-World SOC Application

This visualization would be used to:

- Detect attackers establishing persistence via local admin access
- Monitor for unauthorized privilege escalation attempts
- Track compliance with privileged access management policies
- Investigate insider threat activities involving admin rights